



HANDOFF: Service Pro CRM (clean rebuild)

Resume note for picking this back up after a context clear. **Everything below is committed & pushed.**

II RESUME HERE (2026-07-17b, line-item pricing + optimistic sync)

- **Live deploy confirmed working** by owner (db.gs + setup.gs were the missing pieces; SCHEMA error gone).
- **This session fixed two owner-reported issues in the live app (front-end only, WebApp.html):**
 1. **Estimate line items were stuck at \$0 with no way to set a price.** The builder now has an editable **"Price ea."** field per line that auto-fills from the service's default and is overridable per estimate. The override is sent to the server as `rate` (backend `API_insertLines_` already honored it). Same field added to **Edit line items**.
 2. **New "Price book" screen** (More ▶  Price book) to set default prices for services so they auto-fill estimates; uses the existing `apiListServices/apiCreateService/apiUpdateService/apiArchiveService`.
 3. **Create estimate + Approve are now optimistic** (was a blocking round-trip → slow nav). They write to local state, navigate instantly ("Estimate created" / "Approving..."), and queue the Sheets write in the background via the existing `enqueue` engine (with rollback on failure). Same pattern as clients/jobs. New `refreshBilling()` reconciles server truth while keeping in-flight temps.
- **⚠ To deploy this: re-paste ONLY `WebApp.html`** into the Apps Script project (file shows as `WebApp`). No `.gs` changes; `db.gs/setup.gs/Api.gs` are unchanged, so no re-run of setup and no new web deployment needed (just save the file; hard-refresh the web app). BUILD tag is now `2026-07-17b-lineprice+optimistic`.
- **Tests:** 116 assertions green (db 38 · api 39 · code 15 · web 24). `node tests/webtest.js` covers the price override, optimistic temp + reconcile, and price-book save.
- **Note on \$0 defaults:** `setup.gs` still seeds the 5 services with `DefaultRate: 0` by design (rates are business-specific). The owner sets them once via the new Price book, or overrides per line.

II RESUME HERE (2026-07-17, end of session, deploy in progress)

- **Status:** code complete + all 4 reviews fixed (111 sandbox tests green). Owner is **deploying to the live Apps Script project now**.
- **⚠ IMPORTANT: everything is on branch `claude/revenue-idea-2k-pj7f90`, NOT `master`**. The repo's default branch is `master` and does **not** contain the rebuild. GitHub shows `master` by default, so `db.gs` and `setup.gs` (the two NEW files) look "missing" unless you switch the branch

dropdown to `claudio/revenue-idea-2k-pj7f90` . Not merged to master; no PR opened yet (owner may want one).

- **What tripped us up:** owner ran Set-up and got `ReferenceError: SCHEMA is not defined` ; the cause was copying files from `master` (old app) and missing `db.gs` (where `var SCHEMA` lives). Fix = paste **all 6 files from the branch:** `db.gs` , `setup.gs` , `Api.gs` , `Code.gs` , `WebApp.html` , `Sidebar.html` (+ `appsscript.json`). In the Apps Script file list they show as `db` , `setup` , `Api` , `Code` , `WebApp` , `Sidebar` .
- **Next when resuming:** confirm the live deploy worked: run `setupDatabase()` , walk one flow (Client→Estimate→Approve→Invoice→Paid), and send one test invoice with a logo (PDF check). Then offer to merge the branch → master if the owner wants it on the default branch.
-  **TOP NEXT TASK (owner asked):** the app "loads between pages." Make `WebApp.html` **local-first with** `localStorage` : (1) persist `S.*` (clients/jobs/billing/services/settings/dashboard) + the sync queue to `localStorage` ; (2) on boot render instantly from local, then refresh in the background; (3) replace the broad `invalidate()` (nulls all caches → full reload on next nav) with in-place cache updates. Collisions are NOT a concern (single owner, slow-changing data). The leftover `STORE='crm2_sync'` constant is currently unused. NOTE: branch tip moved to `e7f6ced` outside this session; re-read the current `WebApp.html` before editing.
- **Repo / branch:** `carlvereen/reflections` · branch `claudio/revenue-idea-2k-pj7f90`
- **As of commit:** `8154e6d` (UX review fixes). Working tree clean, level with origin.
- **Date:** 2026-07-17
- **What this is:** a Google Apps Script + Google Sheets CRM for a solo/small home-service business, sold as a packaged product. This session was a **clean rebuild onto a new relational data model** (was name-joined + cell-formula; now **ID-joined, Script-owned, no formulas**), followed by **four reviews** (database, security, code, UX), all fixed.

The app is 6 files (paste each into the Apps Script project)

File	Lines	Role
<code>db.gs</code>	~400	Data layer. <code>SCHEMA</code> registry (single source of truth) + ORM-lite: read-once/write-batch, script lock, <code>insert</code> / <code>insertMany</code> / <code>getById</code> / <code>getAll</code> / <code>query</code> / <code>update</code> / <code>softDelete</code> , <code>recalcDocTotal</code> / <code>recalcClientLifetime</code> , <code>nextId</code> /counter helpers, key/value settings.
<code>setup.gs</code>	~115	<code>setupDatabase()</code> builds all tabs from <code>SCHEMA</code> (headers, freeze, dropdowns, checkboxes, formats, seeds). Also <code>resetDatabase()</code> , <code>verifySchema()</code> .

File	Lines	Role
Api.gs	~477	Web API (doGet + all api*), ID-based, script-computed dashboard, snapshot pricing, PDF/email, logo→Drive. Self-contained API_* helpers.
Code.gs	~232	Menu (onOpen), 5 automations (overdue / recurring roll-forward / follow-up digest / reminders / review requests), daily triggers, Google-Form intake (onFormSubmit), sidebar helpers.
WebApp.html	~563	Mobile SPA. Bottom nav (Home/Clients/Jobs/Billing/More), tabs, fixed action bar, optimistic sync engine, accent theme, logo upload. Built-in preview mock() for standalone render.
Sidebar.html	~72	Desktop Quick Actions (capture a lead, see due follow-ups).
appsscript.json	n/a	Manifest: timeZone (set to owner's tz!), V8, web app access = MYSELF .

Data model (7 tabs + hidden _meta)

Clients (Status = Lead|Active|Inactive|Lost, no separate Leads tab) · Jobs · Estimates · Invoices · LineItems (fastest-growing; polymorphic DocID per DocType) · Services (price book) · Business Settings (key/value: profile, tax, payment, **invoice numbering**, brand) · _meta (ID counters).

Non-negotiables, all in force: join by **ID** never name · **no cell formulas** (Script writes every value) · money=number + \$ via format · dates=real Date · booleans=checkboxes · enums=dropdowns · cached totals Script-owned (Estimates/Invoices.Subtotal/Tax/Total , LineItems.LineTotal , Clients.LifetimeSpent) · **snapshot pricing** (LineItem copies Desc+Rate from Service at write) · **per-doc TaxRate snapshot** · **soft-delete** via Archived · FK+enum validated on write (rejects archived parents; Service refs exempt) · formula/CSV-injection guard on text writes.

Verification (no live Sheet in the build env)

Fake-Sheets/headless sandboxes in tests/ : **111 assertions, all green** (db 38 · api 39 · code 15 · web 19). Run: node tests/dbtest.js / apitest.js / codetest.js / webtest.js from products/service-crm/ . These prove logic/flow, **not** live Apps Script behavior.

Deploy (owner steps)

1. Paste all 6 files into the bound Apps Script project.
2. Reload sheet → ⚡ **CRM** ▶ **Set up / rebuild database** (builds fresh tabs; destructive on existing data).
3. ⚡ **CRM** ▶ **Turn ON daily autopilot** (installs triggers; authorize email/Drive once).
4. **Deploy** ▶ **New deployment** ▶ **Web app** ▶ **Execute as: Me** · **Who has access: Only myself** ▶ **Deploy**.

5. In Apps Script **Project Settings**, set the **time zone** to the owner's locale.

⚠ **Still to confirm on a REAL sheet (the two open items)**

1. Run `setupDatabase()` live and eyeball the tabs + one full flow (Client→Estimate→Approve→Invoice→Paid).
2. **PDF logo**: send one test invoice with a logo set; confirms the Drive-hosted image renders in the Apps Script HTML→PDF converter (the one thing the sandbox can't exercise).

The four reviews (all fixed; see commits a7545e6, a0e1740, 3faf504, 8154e6d)

- **Database**: archive integrity (reject archived-parent FKs; block/guard parent archive), per-doc tax snapshot, auto-recalc lifetime on paid-invoice change, cent rounding, immutable DocType/PK, Invoice→Estimate link, numeric-setting sanitize, timezone manifest.
- **Security**: web-app access `MYSELF` (was `ANYONE_ANONYMOUS`, critical), formula/CSV-injection guard, server-side logo size cap, escaped URLs in PDF/email.
- **Code**: fixed `openDoc` infinite refetch loop, `apiListClients` $O(n^2) \rightarrow O(n)$, optimistic-temp rollback on failure (+queue unblock), silent-failure `.catch` es.
- **UX**: full status control on invoice/estimate detail (un-pay a mis-marked invoice), archive confirmations, live Subtotal+Tax+Total in the estimate builder, due/valid dates, pickClient loads clients first.

Notes / possible next steps (none blocking)

- Old docs in this dir (`PERFORMANCE.md` , `DEPLOY.md` , `SMOKE-TEST.md` , `LISTINGS.md` , marketing HTML) describe the PRE-rebuild app: stale; update if productizing.
- Deferred by design: `Payments` table (partial payments), skipped; `LineItems_Archive` yearly move when the tab nears ~50k rows (~year 3); `apiApproveEstimate` is non-atomic (Sheets has no transactions).
- Wrap-up narrative for the Obsidian vault: `SESSION-WRAPUP-2026-07-17.md` .